🔑 **Answer Key — Teacher Copy · Lesson 2-8**

MSTAR PRACTICE WORKSHEET

Lesson 2-8: Describe Data Using the Mean

Grade 6 Mathematics · Standard 6.DS.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Find the mean of: 4, 6, 8, 10

A. 6

B. 7

C. 8

D. 28

Add the values: 4 + 6 + 8 + 10 = 28. There are 4 values, so the mean is 28 ÷ 4 = 7.

Question 2 SELECTED RESPONSE — CORRECT: B

Miguel earned $60 walking dogs over 4 days. What is his mean daily earning?

A. $4

B. $15

C. $20

D. $60

The total is $60 and there are 4 days, so the mean is 60 ÷ 4 = $15 per day.

Question 3 SELECTED RESPONSE — CORRECT: B

Find the mean of: 3, 5, 7, 9, 11

A 5

B 7

C 9

D 35

Add the values: 3 + 5 + 7 + 9 + 11 = 35. There are 5 values, so the mean is 35 ÷ 5 = 7.

Question 4 SELECTED RESPONSE — CORRECT: A

In her first 4 games, Sofia scored 12, 20, 16, and 11 points. What is her mean points per game so far?

A 14.75

B 15

C 20

D 59

Add the points: 12 + 20 + 16 + 11 = 59. There are 4 games, so the mean is 59 ÷ 4 = 14.75 points per game.

Question 5 SELECTED RESPONSE — CORRECT: D

Sofia wants to average 15 points per game over 10 games. How many points does she need in all 10 games combined?

A 1.5 points

B 15 points

C 25 points

D 150 points

If every one of the 10 games hit the target of 15 points, the total would be 15 × 10 = 150 points. That is the total she needs for a mean of 15.

Question 6 SELECTED RESPONSE — CORRECT: C

Sofia's points in games 1 through 9 were 12, 20, 16, 11, 17, 14, 15, 10, and 18. How many points has she scored so far?

A 15 points

B 123 points

C 133 points

D 150 points

12 + 20 + 16 + 11 + 17 + 14 + 15 + 10 + 18 = 133 points after 9 games.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Find the mean of this data set: 6, 10, 14, 8, 12.

A 8

B 10

C 12

D 50

Sum = 6 + 10 + 14 + 8 + 12 = 50. Mean = 50 ÷ 5 = 10.

- **A:** That is one of the five values, not their average. The sum is 50, and 50 divided by 5 is 10.
- **C:** That is one of the five values, not their average. Add all five and divide by 5.
- **D:** That is the sum of the five values. The mean divides that sum by how many values there are.

Part B — correct answer: D

The mean is often described as a 'fair share' or a balance point. Why?

A The mean is always the middle value when ordered.

B The mean is the value that occurs most often.

C The mean is halfway between the minimum and maximum.

D If the total were redistributed equally among all the values, each would receive the mean.

Adding all values and dividing by the count answers: if this total were shared out evenly, how much would each get? That is the fair-share meaning of the mean.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D, E

Select ALL statements that are true about the mean:

- **A** Mean = sum of all values ÷ number of values.
- **B** Adding one very large value pulls the mean upward.
- **C** The mean must be one of the values in the data set.
- **D** The mean can be a decimal even when every value is a whole number.
- **E** The mean uses every value in the data set.

A, B, D, and E are true. C is false — the mean of 2 and 5 is 3.5, which is not in the set.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

Sofia realized she made an error: she actually scored 12 points in Game 8, not 10. Her other games are unchanged (12, 20, 16, 11, 17, 14, 15, and 18). How does this change the number of points she needs in Game 10 to average 15 points per game? Show how you know.

Model answer: Raising Game 8 from 10 to 12 adds 2 points, so the 9-game total goes from 133 to 135. The target has not changed: a mean of 15 over 10 games still needs 15 × 10 = 150 points in all. So Game 10 must now supply 150 − 135 = 15 points instead of 17 — exactly 2 fewer, because every point gained earlier is a point she no longer has to score later.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.